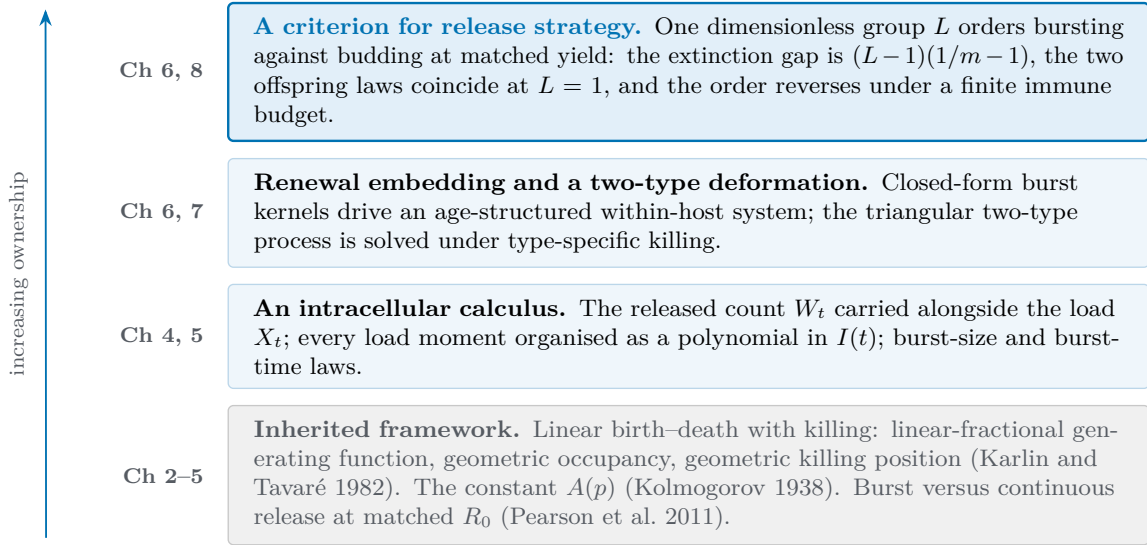


Originality of the results

An index of what is inherited, what is extended, and what is new

Every entry names the published ancestor and the increment made to it, and points to the theorem and page where the result is stated. The analytical spine of this thesis — the linear birth–death process with per-capita total catastrophe — is not new, and is not claimed to be: it is Karlin and Tavaré’s killed birth–death process, and the constant $A(p)$ of Chapter 3 is Kolmogorov’s. Built on that spine are a closed-form intracellular calculus, its embedding in a within-host renewal system, a two-type deformation, and a criterion that orders bursting against budding. Four statements were not found in the literature as written; they come first, each beside the neighbour that comes closest.



1. STATEMENTS NOT FOUND IN THE LITERATURE AS WRITTEN

The method behind each is published; these are original *statements* obtained by specialising it, not a new theory of branching. Each was searched against the chapter bibliographies and the wider literature, then challenged adversarially.

WHERE	THE STATEMENT	NEAREST PUBLISHED WORK
Thm 6.8 p. 163	The flooding identity. At matched lifetime yield the extinction probabilities of bursting and budding differ by exactly $\Delta z_{\text{ext}} = (L - 1)(1/m - 1)$, so bursting favours establishment precisely when $L > 1$, equivalently when $1/\delta < 1/\lambda + 1/\mu$.	Pearson, Krapivsky and Perelson (2011) find $p_v^{\text{burst}} \leq p_v^{\text{cont}}$ — bursting always favoured — from a fixed burst size, with no parameter that can reverse the order.
Prop 6.9 p. 165	Coincidence of the offspring laws. At $L = 1$ the bursting and budding offspring <i>distributions</i> coincide, not merely their extinction probabilities: the burst generating function collapses to the geometric $(1 - \varrho)/(1 - \varrho z)$. The variance gap $2q^2 V_\infty(L - 1)/(a - 1)$ changes sign with $L - 1$.	Pearson’s geometric comparator; Karlin and Tavaré’s geometric killing position. Neither yields a coincidence of laws.

WHERE	THE STATEMENT	NEAREST PUBLISHED WORK
Thm 7.4 p. 189	Exact finite-time survival under two-type killing. A closed form for the no-catastrophe probability of the triangular two-type birth–death process with type-specific catastrophe rate $\delta_1 X + \delta_2 Y$, as a Wronskian mixture of a real-axis ${}_2F_1$ pair.	Antal and Krapivsky (2011) apply the same Gauss reduction to terminal counts without killing. Setting $\delta = 0$ here forces $S \equiv G \equiv 1$ and does <i>not</i> return their generating functions.
Eq. 7.50 p. 201	The killed bivariate generating function. The defective PGF under the same killing, by the same Wronskian mixture. This is the object for which $\delta_i = 0$ <i>does</i> recover Antal–Krapivsky, and it is what makes the previous entry a different observable rather than a reparametrisation.	Antal and Krapivsky (2011), eqs. (25)–(29), recovered here as the $\delta_i \rightarrow 0$ limit.

2. EXTENSIONS: AN ANCESTOR, AND WHAT IS ADDED TO IT

Ordered by chapter. In each case a published result is the ancestor; the thesis changes the process, the object, the hypotheses, or the biological reading.

WHERE	THE INCREMENT	ANCESTOR
§2.2.4.1 p. 35	Gated release in which the opening intensity ηA_i depends on the accumulated load, rather than switching autonomously.	Davis’s piecewise-deterministic Markov processes; autonomous on–off queues.
Thm 3.5 p. 82	$A(p) = 2\psi_{2p}(1/2)$ identifies Kolmogorov’s constant with the Koenigs linearising function of the logistic map, and that function is hypertranscendental — so it admits no elementary closed form.	Kolmogorov (1938) for $A(p)$ itself; Koenigs (1884); Becker and Bergweiler (1995) for hypertranscendence.
Thms 4.13–4.14 pp. 100–101	The released count W_i carried as a second coordinate: its mean, second moment and variance at <i>finite</i> time, as functions of $I(t)$.	Karlin and Tavaré’s killing position is the <i>eventual</i> geometric, not a time-dependent release coordinate.
Prop 4.17 p. 103	Every load moment is a polynomial $Q_p(I)$ of degree $p + 1$ with roots a, b and leading coefficient $(-1)^p p! (\lambda/\delta)^p$ — one economy organising the whole moment hierarchy.	The moments themselves sit inside the 1982 generating function; the organisation is what is added.
Thm 5.10 p. 126	Burst size equals the quasi-stationary law, proved by size-biasing: the weight k makes the burst-time integrand a perfect differential, which telescopes to the endpoint. Remark 5.11 gives a <i>sufficient</i> criterion for it; necessity is left open.	Karlin and Tavaré give the killing-position law and the quasi-stationary law separately, in different lemmas.
Props 5.12–5.13 pp. 130–131	Burst timing: at $\mu = 0$ the time to a burst of size n is harmonic in n with a Gumbel limit; with death, a closed mean conditional on bursting.	Classical Yule maxima, not previously written as a law for a lytic single-cell assay.
§5.9–5.10 pp. 134–147	Multi-founder kernels g_k with sub-extensive yield $k + \lambda/\delta$; a chained immediate-transfer process with negative-binomial rupture sizes and interval Laplace transforms.	The branching property is classical; the yield identity, and the failure of the naive free sum, are the additions.

WHERE	THE INCREMENT	ANCESTOR
Def 6.1, Eq. 6.11 pp. 153–155	An age-structured within-host system driven by <i>closed-form</i> birth–death–catastrophe kernels, with no free production kernel left to fit.	Nelson, Gilchrist, Coombs, Hyman and Perelson (2004), who leave the age-dependent kernel $P(a)$ free.
Thm 6.3 p. 157	A projection of the exponential phase onto an effective pair $(p_{\text{eff}}, d_{I,\text{eff}})$, with closed endpoints — so a fitted burst rate is a property of the phase, not of the cell.	Euler–Lotka renewal theory; Wallinga and Lipsitch on generation intervals.
§6.5.6 p. 161	A concrete three-to-two identifiability obstruction: (λ, μ, δ) is not recoverable from population data alone.	Miao, Xia, Perelson and Wu (2011) on structural non-identifiability.
Prop 7.2 p. 179	A Feynman–Kac representation for the joint two-type occupation hazard, with a mesh-limit argument for the pathwise exponential weight.	Standard first-step decomposition for one type; the joint two-type object is the increment.
Prop 7.9 p. 198	At $\lambda_2 = 0$ the solution is built on <i>ordinary</i> Bessel functions J_ρ, Y_ρ , a different boundary of the parameter space.	Antal and Krapivsky’s bi-critical case uses <i>modified</i> Bessel functions, from a different equation.
Def 8.1, Prop 8.3 pp. 210–212	A replicator–suppressor process in which suppressors are consumed one-to-one and never replenished. The escape wall $i > j$ is then <i>rate-independent</i> : a counting argument, no rates enter.	Pilyugin and Antia (2000) limit clearance by <i>handling time</i> : an engaged effector is occupied, then returns to the free pool. Their threshold therefore depends on the rates.
Prop 8.7, Cor 8.8 pp. 218–219	Under an integer removal budget ϑ , the surviving fraction is $\mathcal{R}(\vartheta) = V_\infty a^{-\vartheta}$ with threshold $\vartheta^* = \log V_\infty / \log a$.	Geometric stop-loss is actuarial folklore; not found for a catastrophe process under an integer immune budget.
Prop 8.12 p. 220	Under threshold removal the flooding criterion <i>reverses</i> : bursting now wins exactly when $L < 1$. The same dimensionless group orders the comparison in both directions.	Yuan and Allen (2011) find the ranking of budding and bursting to depend on whether the immune response is activated, by a different mechanism.
Cor 9.3 p. 233	A two-layer clock: speciation intensity tracks unused <i>individual</i> capacity through a separate logistic abundance layer, not the species count. The limit law is geometric, with a variance not previously recorded.	Kendall (1948) time-dependent Yule; diversity-dependent models act on species number.
Eq. 9.21 p. 240	A closed form for the moving unstable critical point of the Pimentel resistance chain: a reversal is that point crossing the leading population.	Pimentel (1965) supplies the chain; no antecedent was identified for the critical point.

3. INHERITED, AND CITED AS SUCH

Naming these is what makes the two sections above legible as increments. **Karlin and Tavaré (1982)** supply the killed birth–death process itself: the linear-fractional generating function, the closed forms $I(t)$, $D(t)$, $J(t)$, $K(t)$, the roots a, b , the geometric occupancy and the geometric killing position. **Brockwell, Gani and Resnick (1982)** is the more general neighbour, admitting immigration and random catastrophe sizes. **Kolmogorov (1938)** and **Yaglom (1947)** give the constant $A(p)$, the critical decay $S_n \sim 2/n$ and the conditioned limit law; **Pakes (2026)** is the current word on whether that constant admits an explicit form. **Kendall (1948)** gives the linear birth–death generating

function reused throughout, and the time-dependent Yule process of Chapter 9. **Antal and Krapivsky (2011)** give the triangular two-type process and its Gauss reduction. **Perelson (1996)**, **Nowak and Bangham (1996)** and **McLean (1993)** give constant-rate viral replication; **Pearson, Krapivsky and Perelson (2011)** give the invariance of R_0 under matched lifetime yield and the establishment comparison; **Komarova (2007)** gives flooding as a biological idea, and **Yuan and Allen (2011)** a published ranking switch under an immune term. **Nelson et al. (2004)** give the age-structured system; **Miao et al. (2011)** non-identifiability; **Pilyugin and Antia (2000)** finite nonspecific killers. **Nee, May and Harvey (1994)** give the push of the past, against a large diversity-dependent literature. **Ke, Chen and Yang (2013)** give the temperature- and pH-dependent account of the *Yersinia pestis* switch.

4. SCOPE OF THE CLAIMS

“Not found” is not a proof of absence. Where the evidence was thin the claim was weakened rather than strengthened, and every statement in Section 1 has a published ancestor supplying the method — so each is written as an original identity obtained by specialising a known technique. Three things are deliberately *not* claimed: the killed birth–death process and its closed forms, which are Karlin and Tavaré’s; the constant $A(p)$, which is Kolmogorov’s; and the comparison of bursting with budding as a question, which is Pearson’s and Komarova’s. What is claimed is the calculus built on the first, the Koenigs dictionary for the second, and the criterion L that settles the third in both directions.

5. QUICK REFERENCE: WHERE THE CONTRIBUTIONS SIT

CH	PAGES	WHAT IT CONTRIBUTES
1	7–21	Framing, biology and motivation. No result is claimed here.
2	22–57	Toolkit: Markov chains, branching processes, the birth–death–catastrophe process. The increment is gated release with a state-dependent opening intensity (p.35).
3	58–87	Kolmogorov’s constant $A(p)$, recast as the Koenigs linearising function of the logistic map and shown hypertranscendental (Thm 3.5 , p. 82).
4	88–111	The intracellular calculus: the released count W_t as a second coordinate, and every load moment as a polynomial in I (Thms 4.13–4.14 , Prop 4.17).
5	112–147	The complete law of one bursting cell — burst size, burst timing, several founders, chained transfer (Thm 5.10 , Props 5.12–5.13).
6	148–175	Two of the four new statements. The renewal embedding, and the criterion L ordering bursting against budding (Thm 6.8 , Prop 6.9).
7	176–204	Two of the four new statements. Exact finite-time survival under two-type killing, and the killed bivariate generating function (Thm 7.4 , Eq. 7.50).
8	205–227	Two finite-budget immune caricatures, and the reversal of the flooding criterion under threshold removal (Prop 8.12).
9	228–256	A two-layer capacity clock on a Yule process (Cor 9.3) and a moving critical point in the Pimentel chain (Eq. 9.21).
10	257–272	Synthesis, and a consistency check on the <i>Yersinia pestis</i> release strategy (Eq. 10.9 , p. 263) offered explicitly as a check rather than a proof.